

ORIGINAL ARTICLE

Oxytocin neural responses distinguish social novelty from familiarity but not kin from non-kin in male spiny mice

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Funding information

National Science Foundation, Grant/Award
Number: IOS-2310626; Department of
Defense through the National Defense
Science and Engineering Graduate Fellowship
Program

Abstract

In most species, individuals must be able to identify threats, peers, and potential mates to survive. The distinction of kin from non-kin and novel conspecifics from familiars is essential to the successful categorization of these identities. Although oxytocin (OXT) signaling has been implicated in social recognition, little is known about the contributions of distinct OXT-producing cell groups to distinguishing conspecific type. To determine whether OXT-producing neuronal populations differentially respond to novelty or kinship status, we conducted immediate early gene tests in male spiny mice (*Acomys dimidiatus*), a communally breeding species that we previously showed distinguishes between novelty and kinship status. Immunohistochemical analysis of brain tissue revealed that the OXT cell populations in the paraventricular nucleus of the hypothalamus and the anterior hypothalamus did not differentially respond to the kinship or novelty status of same-sex conspecifics. However, while OXT-producing neurons in the bed nucleus of the stria terminalis did not distinguish between kin and non-kin spiny mice, this cell group was more responsive to familiar than novel conspecifics. These results suggest that extrahypothalamic OXT neurons may be involved in aspects of processing the novelty status of a conspecific.

KEYWORDS

bed nucleus of the stria terminalis, kinship, novelty, oxytocin, social recognition

1 | BACKGROUND

Social recognition and the ability to distinguish distinct types of conspecifics is crucial for all animals. Incorrect assessments about who is friend or foe could potentially be fatal, and a failure to determine who is closely related kin could yield unviable offspring. For communally breeding species that live in large groups and experience an influx of new members due to dispersal from neighboring groups, the ability to distinguish between different types of conspecifics may be especially important. For example, accurate identification of whether or not an opposite-sex new group member is a half-sibling can have serious fitness consequences. Because interactions with different types of conspecifics can convey varying benefits and consequences, the brain has evolved mechanisms to dissociate conspecifics based on various

identity characteristics. Indeed, we recently showed that the lateral septum exhibits differential neural responses to kinship status, but not novelty status in colonial spiny mice.¹ These findings are similar to those from studies in Long-Evans rats, which have identified a neural topographic mapping of kinship in the lateral septum.² Lateral septal circuitry via connections with the hippocampus also mediates recognition and preferences for social novelty in C57BL/6J mice.³ In addition to identifying brain regions and neural circuits involved in social recognition, we can refine our understanding of neural mechanisms underlying this cognitive task by examining specific cell types that modulate the discrimination of social stimuli.

Oxytocin (OXT) signaling is a potent mediator of numerous types of social behavior, with several studies demonstrating an involvement in processing distinct types of social information. For example, global

knockout of the OXT gene or OXT receptor (OXTR) gene impairs social memory and novel-familiar discrimination in mice.^{4–6} Studies that have examined site-specific contributions of OXT signaling to social recognition have demonstrated that hippocampal OXTRs are necessary for discriminating novel from familiar same-sex conspecifics in male C57BL/6J mice,⁷ activation of OXTRs in the anterior olfactory nucleus enhances novelty detection of same-sex conspecifics in Wistar rats,⁸ and blockade of OXTRs in the bed nucleus of the stria terminalis (BNST) impairs the ability to distinguish novel from familiar same-sex conspecifics in Wistar rats.⁹ While most studies examining social recognition focus on novel-familiar discrimination, one study in humans revealed a role for OXT specifically in kin recognition and found that administration of intranasal OXT enhances kinship recognition (i.e., recognition of faces of close familial relationships) in women, but not men.¹⁰ The neural mechanisms by which OXT signaling may influence kin recognition are, to our knowledge, largely unknown. Further, while several studies have examined OXTR-mediated social recognition, we know far less about the contributions of OXT source populations to social discrimination abilities.

OXT-producing neuronal populations can be found throughout the basal forebrain and midbrain, with the largest population being located in the paraventricular nucleus of the hypothalamus (PVN) and smaller hypothalamic populations located in the supraoptic (SON) and anterior (AH) hypothalamic regions, along with scattered OXT cells found throughout the lateral hypothalamus.^{11–13} The PVN and SON OXT neuronal populations modulate basic physiology and stress response via projections to the posterior pituitary.^{14,15} PVN OXT cells also send projections throughout the brain^{14,16} to modulate a variety of social behaviors, including the promotion of affiliation in CD-1 mice and prairie voles.^{17,18} One study has also demonstrated a role for the PVN OXT cell group in social recognition, such that chemogenetic inhibition of these neurons impairs the ability to discriminate between novel and familiar same-sex conspecifics in male Sprague Dawley rats.¹⁹ Few studies have directly manipulated AH OXT neurons; however, infusion of OXT into the AH of female Syrian hamsters decreases aggression,²⁰ suggesting that this cell group may have pro-social functions. The most evolutionarily conserved extrahypothalamic OXT cell group is found in the bed nucleus of the stria terminalis (BNST)—a cell population that facilitates social vigilance in California mice²¹ and thus may also play a role in directing attention to specific social cues.

In the present study, we sought to determine whether distinct OXT neuronal populations distinguish between novel and familiar, as well as kin and non-kin social stimuli. To do this, we used spiny mice (*Acomys dimidiatus*), a highly social and communally breeding rodent native to the Middle East, Africa, and Southeastern Asia.²² Spiny mice preferentially associate with larger groups and readily accept newcomers into established groups.^{23–25} In the wild, they live in groups ranging from 12 to 60+, with overlapping territories of other *Acomys* species,^{26,27} suggesting that *Acomys* has evolved to frequently encounter a variety of conspecific types. The precise mechanism used by spiny mice for the discrimination of these conspecific types remains elusive; yet spiny mice do show phenotype matching,²⁸

suggesting that perhaps kinship and novelty recognition are likely to involve odor cues just like in other rodent species.²⁹ Regardless, we have previously demonstrated that spiny mice can distinguish between novel and familiar conspecifics in a habituation-dishabituation test³⁰ and recently found that male spiny mice can also distinguish the kin status of conspecifics—a phenomenon reflected in the differential responsiveness of neurons in the lateral septum.¹ In this experiment, we examined the behavior of male spiny mice that freely interacted with a novel same-sex group comprised of familiar kin, novel kin, and novel non-kin. Males preferentially affiliated with novel kin, indicating an ability to distinguish the kin status of same-sex conspecifics in a complex group setting. Further, in dyadic interactions, males exhibited more pro-social behavior with a novel kin conspecific than with a novel non-kin conspecific.¹

To examine PVN, BNST, and AH OXT neural responses to conspecific type, we conducted two immediate early gene (IEG) studies. For the first IEG study, to examine neural responses to novel vs. familiar conspecifics while controlling for kinship status, we exposed male spiny mice to either a novel, non-kin same-sex conspecific or a familiar, non-kin same-sex conspecific. For the second IEG study, to examine neural responses to kin vs. non-kin conspecifics while controlling for novelty status, we exposed male spiny mice to either novel, same-sex kin or novel, same-sex non-kin. Because the inhibition of OXT neurons in the PVN impaired novelty recognition in rats,¹⁹ we hypothesized that this cell group would be more responsive to novel than familiar conspecifics in spiny mice. Furthermore, because the lateral septum is involved in kin recognition in spiny mice¹ and rats² and because the BNST sends direct axonal projections to the lateral septum,³¹ we hypothesized that BNST OXT neurons would be differentially responsive to kin vs. non-kin exposure.

2 | METHODS

The behavioral data and brain tissue used for this experiment are from a study that was previously published in Scientific Reports.¹ This former study quantified neuronal responses in the lateral septum, whereas the present data reflect new analyses from data obtained through new immunohistochemical staining of a previously unexamined series of tissue. Thus, the neural data and brain-behavior relationships presented in this study are original.

2.1 | Animals

All animals were obtained from our breeding colony, which originates from the captive-bred colony of Dr. Ashley W. Seifert at the University of Kentucky. Thirty-two adult male spiny mice, *Acomys dimidiatus*, age post-natal day (PND) 60 to 512 were used for the immediate early gene (IEG) study. Sex was determined by examination of gonads. Spiny mice were housed at an ambient temperature of $24 \pm 2^\circ\text{C}$ on a 14-h light: 10-h dark cycle. Animals were housed with anywhere from 2 to 4 conspecifics in either a single-level ($40.64 \times 20.32 \times 20.32$ cm)

or two-level (32 × 38 × 40 cm) rat polycarbonate cage. The floor of the cages was covered with Sani-Chip bedding, and spiny mice had unrestricted access to food and water. For enrichment, cages contained rodent igloos, shepherd shacks, and nesting materials. All procedures were approved by the Institutional Animal Care and Use Committee of Emory University. Due to the limited availability of female spiny mice in our colony, we were unable to include them in this experiment, and thus future studies are required to determine the involvement of OXT neurons in distinguishing conspecific type in female spiny mice.

2.2 | Immediate Early Gene Studies

To examine OXT neural responses to distinct types of conspecifics, 32 male spiny mice underwent one of two IEG studies: (1) Kin vs. Non-kin IEG: males interacted with either a novel, same-sex kin conspecific or a novel, same-sex non-kin conspecific or (2) Novel vs. Familiar IEG: males interacted with either a novel, same-sex, non-kin conspecific or a familiar, same-sex, non-kin conspecific. The first IEG study controlled for novelty status, whereas the second IEG study controlled for kinship status; this allows us to better distinguish between neural responses to conspecific type.

For the Kin vs. Non-kin IEG study, novel kin stimuli were siblings of subjects that were born at least two litters before or after the subject, ensuring that the subject had never interacted with or seen the stimulus animal prior to the IEG study. For the Novel vs. Familiar IEG study, familiar non-kin stimulus animals became familiar to the subject by housing the stimulus animal and subject in a large two-level cage with a clear Plexiglas divider that contained 1 cm holes to allow for tactile and olfactory communication. Two subjects (i.e., siblings) were housed on one side, and 2 non-kin conspecifics were housed on the other side of the divider for 1 week prior to testing so that no subjects or stimulus animals were single housed.

For the IEG tests, the subject and stimulus animal were simultaneously transferred via plastic beakers into a standard rat polycarbonate cage. Depending on the IEG test (i.e., Kin vs. Non-kin or Novel vs. Familiar), same-sex stimulus animals could have been: novel kin, novel non-kin, or familiar non-kin. The subject and stimulus animal were allowed to freely interact for 30 min, the stimulus animal was then removed for an additional 30 min, and finally the subject was immediately perfused to capture Fos expression in response to the interaction with the stimulus animal. Social interactions were video recorded from a top-down view using a Sony Handycam camera, and the first 5 min of time spent engaging in prosocial, investigative, and non-overt behavior was quantified using BORIS.³² Prosocial behavior was defined as allogrooming, positive side-by-side contact, and huddling. Investigative behavior was defined as the nose or paws of the subject making contact with the head, flank, or rear of the stimulus animal. Non-overt behavior was defined as the subject spending time not engaging with the conspecific; we categorized all such behavior as “non-overt” rather than “non-social” because the mice may have been communicating via olfactory signals, which we would

have been unable to detect. Behavioral data was reported in our previous study.¹

2.3 | Histology and Immunohistochemistry

Following the IEG studies, mice were euthanized using a lethal dose of isoflurane before undergoing cardiac perfusion. Subjects were first perfused with 0.1 M Phosphate Buffer Saline (PBS) followed by 4% paraformaldehyde (pH 9.5). Brains were promptly removed and post-fixed overnight in 4% paraformaldehyde at 4°C. Afterward, the tissue was cryoprotected through exposure to 30% sucrose dissolved in PBS for 48 h at 4°C. Brains were then transferred into molds containing Tissue-Tek® Optimal Cutting Temperature Compound and frozen at −80°C. Using a Leica cryostat, tissue was sectioned coronally at 40 μm, with every third section saved for this study.

Tissue sections containing the PVN, AH, and BNST were stained for OXT and Fos. The tissue was rinsed on a shaker 5 times for 5 minutes in 0.1 M PBS. Tissue was then transferred to a block solution (10% normal donkey serum, 0.3% triton, and 0.1 M PBS) and stored in a humid chamber at room temperature for 1 hour. Afterwards, the tissue was incubated in primary antibodies diluted in 0.1 M PBS, 5% donkey serum, and 0.3% triton at 4°C in a humid chamber. The primary antibodies used were rabbit anti-OXT (Abcam, 1:250) and mouse anti-Fos (Santa Cruz, 1:500). Following 48 h of incubation in the primary antibody solution, tissue was removed and rinsed 2 times for 30 minutes each in 0.1 M PBS. Sections were then transferred into a secondary antibody solution and stored in the dark in a humid chamber at room temperature for 2 h. The secondary antibodies used were donkey anti-rabbit 488 (3:1000) and donkey anti-mouse 594 (4:1000) diluted in 5% donkey serum, 0.3% triton, and 0.1 M PBS. Lastly, tissue underwent final rinses in 0.1 M PBS and was stored in a humid chamber at −20°C for subsequent mounting of tissue onto slides. Tissue sections were mounted on pre-subbed microscope slides (Trubond 380 Adhesion Slides) and cover slipped with Prolong Gold antifade mounting media containing a DAPI nuclear stain (ThermoFisher Scientific).

2.4 | Neural Quantification

Photomicrographs were obtained using a Zeiss Axiolmager II microscope fitted with an Apotome optical dissector (Carl Zeiss Inc., Jena, Germany). Composite images were acquired using the z stack function of the Zen Pro software (Carl Zeiss Inc., Jena, Germany). The neuroanatomical location of structures was based on Paxinos and Franklin's mouse brain atlas.³³ All cell counts were conducted using FIJI.³⁴

For PVN OXT-Fos cell counts, we quantified the number of OXT neurons and OXT-Fos colocalized neurons from 1 rostral section and 1 caudal section of the PVN, with roughly 240 μm in between the two tissue sections sampled. Cell counts across the sections were totaled, and we calculated the percentage of OXT-ir neurons that were Fos-ir+. Because the SON is densely populated with both OXT

cell bodies and fibers, we did not quantify this structure as it is difficult to produce accurate colocalized cell body counts.

For BNST and AH cell counts, we quantified the number of OXT and OXT-Fos colocalized neurons across three consecutive tissue sections. Cell counts across the sections were totaled, and we calculated the percentage of OXT-ir neurons that were Fos-ir+. Notably, tissue samples from a subject in the Kin vs. Non-kin IEG exhibited significant damage, so analyses for the AH contained one less data point for kin ($n = 7$) as opposed to non-kin ($n = 8$).

2.5 | Statistical Analysis

Behavioral and neural data were analyzed using IBM SPSS Statistics version 29. Outliers were identified utilizing z-scores; 1 datapoint was removed from the linear regression analyses as it was two standard deviations outside the mean for both OXT-fos colocalization percentage and behavior removed from analyses. Additionally, data from 1 subject were not included for the AH OXT cell group due to tissue damage. Neural responses were not normally distributed and were analyzed using Mann-Whitney U tests. Brain-behavior relationships were analyzed using linear regressions and corrected for multiple comparisons using the false discovery rate (FDR) correction.

3 | RESULTS

3.1 | OXT neuronal population responses to kinship status

To determine whether OXT neurons in the PVN, AH, and BNST differentially respond to the kinship status of novel, same-sex conspecifics, for each cell group, we quantified the percentage of OXT cells colocalized

with Fos. Mann-Whitney U tests revealed no significant differences in OXT neural responses between males exposed to novel kin and males exposed to novel non-kin for the PVN OXT (Wilcoxon $W = 58.00$, $Z = -1.050$, $p = 0.294$; Figure 1A), AH OXT (Wilcoxon $W = 55.00$, $Z = -1.050$, $p = 0.298$; Figure 1B), or BNST OXT (Wilcoxon $W = 58.00$, $Z = -1.050$, $p = 0.294$; Figure 1C) cell groups. Together, these findings suggest that the OXT neuronal populations of the PVN, AH, and BNST do not distinguish kinship status in male spiny mice.

3.2 | OXT neuronal population responses to novelty status

To determine whether OXT neurons in the PVN, AH, and BNST differentially respond to the novelty status of non-kin, same-sex conspecifics, for each cell group we again quantified the percentage of OXT cells colocalized with Fos. Mann-Whitney U tests revealed no significant differences in neural responses between males exposed to novel non-kin and males exposed to familiar non-kin for the PVN OXT (Wilcoxon $W = 57.00$, $Z = -1.155$, $p = 0.248$; Figure 2A) or AH OXT (Wilcoxon $W = 53.50$, $Z = -1.524$, $p = 0.128$; Figure 2B) cell groups. However, analysis of the BNST OXT population showed that spiny mice exposed to novel conspecifics exhibited a significantly lower percentage of OXT-Fos colocalization compared to spiny mice exposed to familiar conspecifics (Wilcoxon $W = 48.50$, $Z = -2.049$, $p = 0.040$; Figure 2C). This suggests that BNST OXT neurons may distinguish between novel and familiar conspecifics.

3.3 | Brain-Behavior Relationships

As previously reported in Fricker et al.,¹ in the Kin vs. Non-kin IEG study, males exhibited more prosocial behavior with novel kin than

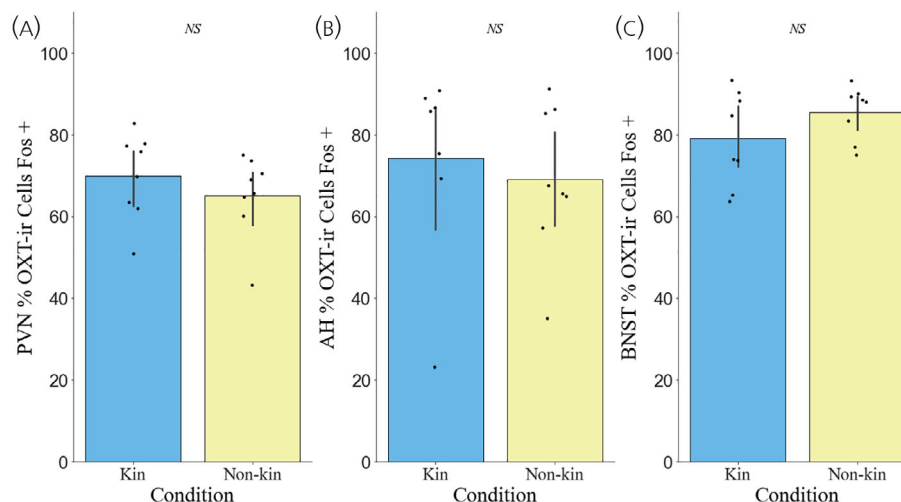


FIGURE 1 Oxytocin neural responses to kinship status. The percentage of OXT-ir neurons that were Fos-ir+ in the (A) PVN, (B) AH, and (C) BNST of spiny mice exposed to either novel kin (sky blue) or novel non-kin (light yellow). NS indicated not significant. Dots represent individual data points. Data are represented as mean ± SEM.

FIGURE 2 Oxytocin neural responses to novelty status. The percentage of OXT-ir neurons that were Fos-ir + in the (A) PVN, (B) AH, and (C) BNST of spiny mice exposed to either a novel (pink) or familiar (orange) conspecific. NS indicated not significant. * indicates $p < 0.05$. Dots represent individual data points. Data are represented as mean $\pm$ SEM.

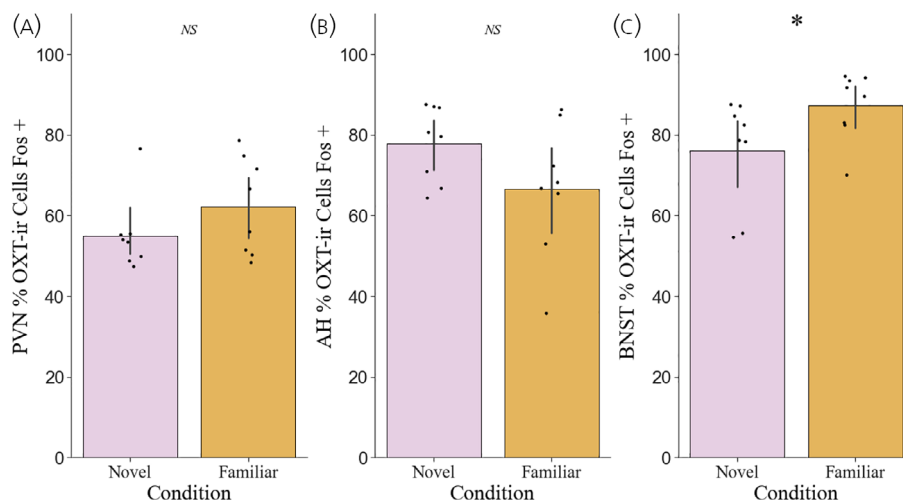


TABLE 1 Statistics from linear regressions examining the relationship between the time spent exhibiting prosocial behavior and the percentage of OXT neurons expressing Fos for the PVN, AH, and BNST.

Conspecific	Explanatory variable	F	df	p-value	R ²	FDR adjusted p-value
Familiar non-kin (N = 8)	PVN	0.858	1,6	.390	0.125	.788
	AH	0.245	1,6	.638	0.039	.788
	BNST	0.163	1,6	.700	0.026	.788
Novel Kin (N = 6)	PVN	0.083	1,4	.788	0.020	.788
	AH	0.385	1,4	.569	0.088	.788
	BNST	3.676	1,4	.128	0.479	.576
Novel non-kin (N = 16)	PVN	0.258	1,14	.619	0.018	.788
	AH	3.718	1,14	.074	0.210	.576
	BNST	0.979	1,14	.339	0.065	.788

with novel non-kin. Further, in the Novel vs. Familiar IEG study, we found that males differentially investigated novel and familiar non-kin conspecifics. To examine whether OXT neuronal responses are predictive of prosocial or investigative behaviors exhibited during interactions with conspecifics, we ran linear regression models with the percentage of OXT-Fos colocalization for each brain area as the independent variable and behavior as the dependent variable.

Because there was an exposure condition in each IEG study in which subjects interacted with novel non-kin, to increase statistical power, we combined these two groups for analysis. Three regressions were run for each conspecific exposure type (i.e., novel kin, novel non-kin, familiar non-kin), one for each OXT cell group (i.e., PVN, AH, BNST), totaling 9 linear regressions per behavior; to correct for multiple comparisons, we used a Benjamini-Hochberg FDR correction. Analyses did not yield any significant relationships between prosocial behavior and the percentage of OXT-Fos colocalization in the PVN, AH, or BNST for animals exposed to either novel kin, novel non-kin, or familiar non-kin (all $p > 0.576$; Table 1). Similarly, we observed no significant relationships between OXT neural responses and non-overt behavior, which was defined as any non-socially interactive behavior (all $p > 0.477$; Table 2).

For investigative behavior, linear regressions showed no significant relationships between investigation and PVN OXT or AH

OXT neural responses (all $p > 0.347$; Table 3). However, we observed a relationship between investigation and BNST OXT neural responses for males that interacted with novel non-kin ($F_{(1,14)} = 6.173$, $p = 0.026$, $R^2 = 0.306$; Figure 3A); yet, this effect became non-significant after FDR correction (FDR Adjusted $p = 0.234$). These data suggest that PVN, AH, and BNST OXT neural responses may not relate to prosocial, investigative, or non-overt behaviors in male spiny in the contexts tested here. However, it is worth noting that we may be statistically underpowered to detect significant relationships if they exist.

4 | DISCUSSION

Here we show that the hypothalamic populations of PVN and AH OXT cells do not exhibit different neural responses to the kinship or novelty status of a conspecific in male spiny mice. However, we did find that BNST OXT neurons are less responsive to novel non-kin than familiar non-kin conspecifics, suggesting that the BNST OXT cell group may play a role in processing the novelty status of an individual.

TABLE 2 Statistics from linear regressions examining the relationship between the time spent exhibiting non-overt behavior and the percentage of OXT neurons expressing Fos for the PVN, AH, and BNST.

Conspecific	Explanatory variable	F	df	p-value	R ²	FDR adjusted p-value
Familiar non-kin (N = 8)	PVN	0.858	1,6	.390	0.125	.788
	AH	0.245	1,6	.638	0.039	.788
	BNST	0.163	1,6	.700	0.026	.788
Novel Kin (N = 6)	PVN	0.083	1,4	.788	0.020	.788
	AH	0.385	1,4	.569	0.088	.788
	BNST	3.676	1,4	.128	0.479	.477
Novel non-kin (N = 16)	PVN	0.406	1,14	.534	0.028	.788
	AH	2.215	1,14	.159	0.137	.477
	BNST	2.571	1,14	.131	0.155	.477

TABLE 3 Statistics from linear regressions examining the relationship between the time spent exhibiting investigative behavior and the percentage of OXT neurons expressing Fos for the PVN, AH, and BNST.

Conspecific	Explanatory variable	F	df	p-value	R ²	FDR adjusted p-value
Familiar non-kin (N = 8)	PVN	0.399	1,6	.551	0.062	.620
	AH	2.189	1,6	.189	0.267	.347
	BNST	0.662	1,6	.447	0.099	.620
Novel Kin (N = 6)	PVN	3.634	1,4	.129	0.476	.347
	AH	0.266	1,4	.633	0.250	.633
	BNST	0.476	1,4	.528	0.106	.620
Novel non-kin (N = 16)	PVN	2.795	1,14	.117	0.166	.347
	AH	1.869	1,14	.193	0.118	.347
	BNST	6.173	1,14	.026	0.306	.234

4.1 | BNST OXT neuronal contributions to social behavior

A previous study reported that social recognition is modulated by OXT signaling in the BNST, such that infusion of OXT into the BNST of male Wistar rats enhances social recognition of novel juveniles in the presence of a familiar juvenile.⁹ This finding is somewhat consistent with our result that BNST OXT neurons differentially respond to the novelty status of a conspecific. However, behavioral results from an infusion of OXT into the BNST reflect the effect of OXT binding at OXT or vasopressin receptors in the BNST. In our study, we specifically assessed BNST OXT-producing neurons, which may release some OXT into the BNST to bind locally but may also send OXT to distant sites via axonal projections to other brain regions, such as lateral and anterior hypothalamic nuclei.²¹ Further, there are important methodological differences between the study in rats and our study here in spiny mice. In the paper by Dumais et al., rats were simultaneously exposed to both familiar and novel juvenile same-sex conspecifics, and social recognition was measured as the percent of time spent investigating the novel juvenile.⁹ This paradigm is more similar to a social preference test, in which an animal has a choice for where it spends its time. Thus, it is feasible that their findings showed that OXT may have facilitated a preference for the novel subject, as

opposed to enhancing social discrimination or recognition, *per se*. Notably, our paradigm was also not designed to specifically test social recognition but rather to examine if distinct OXT neuronal populations differentially respond to novel or familiar conspecifics. Although our findings suggest that BNST OXT may be involved in discrimination processes, follow-up studies are required to determine if BNST OXT neurons directly facilitate social recognition in spiny mice. However, because we exposed spiny mice to a single conspecific, we can rule out the possibility that our findings indicate that BNST OXT neural responses reflect the modulation of a social preference for familiar or novel conspecifics. Instead, the differences we observed in BNST OXT neural responses are more likely to reflect the encoding of specific aspects of the identity of the stimulus animal.

In rodents, the BNST receives olfactory information relayed through the medial amygdala.³⁵ Additionally, studies have demonstrated that OXT can have neuromodulatory effects on olfactory signal processing.^{36,37} Further, OXTRs in the BNST are crucial for the recognition of olfactory cues in Syrian hamsters.³⁸ It is possible that the higher BNST OXT neural responses that we observed in male spiny mice exposed to familiar conspecifics may reflect olfactory processing and/or aspects of social memory. As previously reported, we found that spiny mice from the present study differentially investigated bodily locations of novel and familiar conspecifics.¹ Specifically,

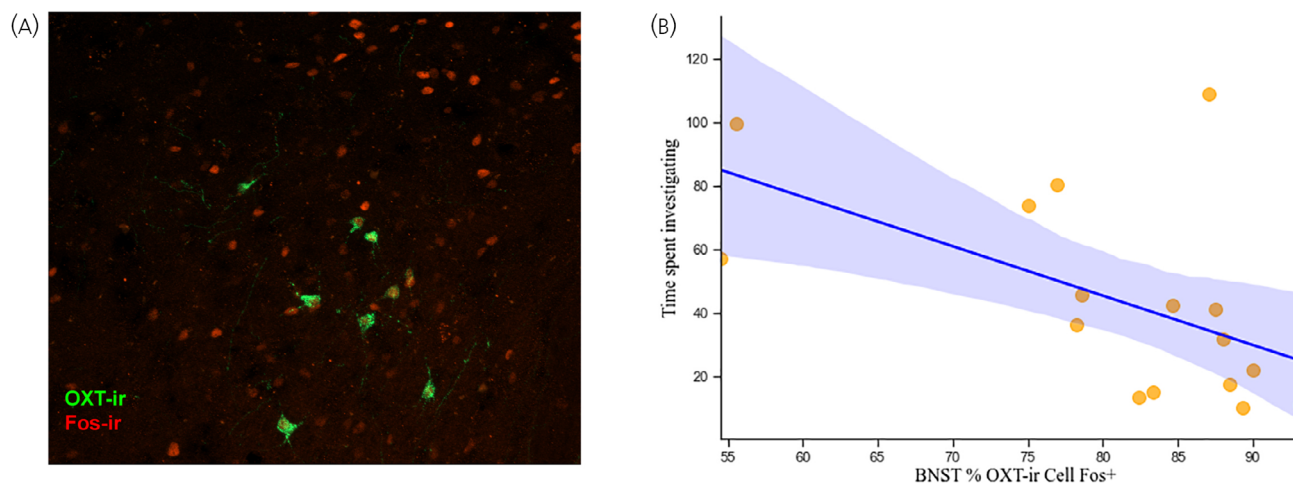


FIGURE 3 BNST oxytocin relationship to behavior. (A) Representative photomicrograph of OXT-ir (green) and Fos-ir (red) labeling in the BNST of a spiny mouse. (B) Time spent investigating a novel non-kin conspecific regressed onto the percentage of BNST OXT-Fos colocalized cells ($n = 16$, $R^2 = 0.31$, unadjusted p -value = 0.026, FDR-adjusted p -value = 0.234).

males investigated the flank more than the head of a familiar conspecific, whereas males that investigated a novel conspecific did not preferentially investigate any specific body region.¹ If BNST OXT neurons are involved in processing olfactory information, this may provide a potential explanation for the significant difference we observed in neural responses between males exposed to novel vs. familiar individuals. Spiny mice possess sebaceous glands in their dermis (across all their skin)³⁹ and harderian glands behind their eyes,⁴⁰ which are known to allow for the release of pheromones and chemical signals in rodents closely related to spiny mice like the Mongolian gerbil.^{41,42} While we have observed glandular tissue in the flank region of male spiny mice (unpub. obs.), to our knowledge, flank glands have not been systematically characterized in spiny mice. Regardless, our behavioral findings indicate that spiny mice gather distinct types of social information via different areas of the body.¹ Because the males in this study differentially investigated familiar conspecifics, it is possible that the greater response of BNST OXT neurons to exposure to a familiar conspecific may reflect olfactory-related processes.

The majority of research examining BNST OXT neuronal function comes from studies using the highly territorial California mouse. In California mice, the BNST OXT cell group is sensitive to social defeat stress (SDS). Male California mice exposed to SDS have greater BNST OXT-Fos colocalization immediately after a third bout of SDS, but not 2 weeks later,⁴³ suggesting these cells are acutely responsive to stress. Similarly, data from studies using prairie voles also suggest that BNST OXT cells may be sensitive to stressors given that fathers separated from their pups exhibit greater OXT-Fos colocalization in the BNST compared to fathers that were reunited with their offspring.⁴³ Together, these findings suggest that BNST OXT neurons respond to stress. However, in the present study, we found that BNST OXT neurons were more responsive to interactions with a familiar conspecific than a novel conspecific; a familiar conspecific, that spiny mice prosocially interacted with, is presumably not more stress-inducing than a

novel conspecific. One interpretation could be that our findings simply do not reflect a stress response. Alternatively, it is important to also consider that OXT has primarily been associated with anxiolytic effects in rodents (i.e., OXT is released to help an animal cope with stress); notably, most observations of anxiolysis come from examination of PVN OXT cells.⁴⁴ Yet, if BNST OXT is released in the face of a stressor to aid in stress-coping, then perhaps a familiar peer—through social buffering mechanisms—could stimulate an increase in OXT release to help the stressed animal to better cope with the stressor. Although purely hypothetical, if this were the case in our study, then that would suggest that the conspecific stimuli (novel or familiar) may not have been the stressors for subjects but rather the spiny mice were stressed due to other factors, such as being in a novel testing cage or being removed from their home cage. Thus, stress levels may have been equivalent across subjects interacting with novel and familiar conspecifics, and it is feasible that BNST OXT cells were more responsive in the presence of a familiar conspecific because that type of conspecific may have been comforting to a highly social animal like a spiny mouse. Indeed, studies in prairie voles have demonstrated that OXT plays an important role in social buffering under stressful conditions, and PVN OXT release mitigates anxiety-like responses.⁴⁵ Notably, if a familiar conspecific induced a greater OXT cellular response in the BNST of spiny mice, potentially providing some degree of stress/anxiety relief for subjects, such a response was not reflected in the amount of time the subject engaged in prosocial behavior with the conspecific; although spiny mice were more prosocial toward novel kin than novel non-kin, spiny mice did not differ in the amount of prosocial behavior exhibited toward familiar or novel non-kin.¹ Because we cannot conclusively determine whether stress played a role in BNST OXT neural responses to familiar or novel conspecifics, future studies are required to determine whether and how BNST OXT responds to social stressors in spiny mice and if this cell group contributes toward regulating, perhaps indirectly, the HPA axis.

4.2 | Hypothalamic OXT neuronal contributions to social discrimination

In the present study, we did not observe differences in PVN OXT responses when spiny mice were exposed to conspecifics that varied in novelty or kinship status. This finding contradicts previous research that showed that social recognition memory of male Sprague Dawley rats is impaired after chemogenetic silencing of the PVN OXT cell group.¹⁹ However, the study exposed male Sprague Dawley rats simultaneously to familiar and novel juvenile rats,¹⁹ and thus differences in time spent with novel and familiar stimuli after chemogenetic silencing could have reflected a modification of social preference rather than social recognition. We previously found that OXT neurons in the PVN exhibit greater responses to social than nonsocial stimuli in spiny mice,²³ suggesting these neurons distinguish between social and nonsocial contexts. If the PVN OXT cell group primarily differentiates between social and nonsocial contexts, at least in spiny mice, then this could explain the lack of significant differences observed here because all subjects were exposed to a social context. Alternatively, PVN OXT neurons may be involved in distinguishing between conspecific types, but we simply lack the cellular resolution to observe such a finding in the present dataset. The PVN contains the largest nonapeptide-producing neuronal populations in the brain and has diverse outputs.⁴⁶ In rats and mice, OXT outputs from the PVN to the amygdala mediate recognition of social cues,^{6,47} and PVN OXT projections to the hippocampus are implicated in social memory and discrimination.^{3,7,48} Because the PVN OXT cell group is rather large, with multiple axonal projections throughout the brain, quantification of the entire cell group may have obscured differences in specific subpopulations of OXT neurons that are differentially responsive to novelty and/or kinship status.

As stated above, the PVN OXT cell group is involved in the modulation of the stress response and has anxiolytic effects on behavior.⁴⁹ Because we did not observe any differences in PVN OXT neural responses to kinship or novelty status, it is possible that interactions with these different types of conspecifics were not stressful for spiny mice or that all subjects experienced roughly equivalent amounts of stress. Interestingly, a recent study using C57BL/6 backcrossed Oxttm1.1(Cre)Doln/J mice demonstrated that nonsocial stressors such as restraint and tail suspension cause greater changes in fluorescence in PVN OXT neurons than exposure to and interactions with social stimuli.⁴⁸ Further, the activity of these neurons during aggressive encounters (i.e., presumably stressful interactions) did not significantly differ from activity during prosocial interactions with conspecifics, suggesting that the PVN OXT cell group may be particularly sensitive to nonsocial stressors.⁴⁸ In our study, any nonsocial stressors would have been consistent across treatment groups, which could explain our null results. Alternatively, similar to our discussion in the previous paragraph, it is possible that global quantification of the PVN OXT cell group masked any differential responses exhibited specifically by magnocellular OXT neurons that send direct axonal projections to the pituitary to modulate the stress response.

Unfortunately, little is known about the role of AH OXT neurons in social behavior. Studies have shown that OXTR activation in the

medial preoptic-anterior hypothalamus (MPOA-AH) continuum of highly aggressive female Syrian hamsters reduces aggression and increases nonsocial behavior towards an intruder, suggesting that AH OXT may counteract AH vasopressin-induced aggressive behavior.²⁰ In a highly social species, AH OXT could serve to both counteract AH vasopressin-induced aggression as well as potentially facilitate prosocial behavior; however, this hypothesis remains untested. Interestingly, spiny mice possess a robust OXT population in the AH¹² and do not exhibit much aggression toward novel individuals.²⁷ Thus, this cell population may support the gregarious behavioral phenotype of spiny mice by suppressing aggression. If this is the case, the lack of differential responses in the AH OXT cell group to different conspecific types in the present study may reflect the lack of aggression exhibited by all subjects.

5 | CONCLUSION

Here, we found that although the PVN and AH OXT cell groups do not distinguish between novelty and kinship status, BNST OXT neurons respond differently to novelty, but not kinship, status in male spiny mice. This finding suggests that BNST OXT neurons are more sensitive to familiar than novel social stimuli. However, it remains unknown whether this greater responsiveness reflects olfactory processing and/or social memory or social buffering in the face of mild stress caused by being a subject in an experiment. While future studies are required to determine the specific aspects of familiar conspecifics that induce a greater BNST OXT neural response, these results demonstrate that at least some OXT neuronal populations contribute toward the differentiation of conspecific type—a characteristic that, to date, has largely been examined only for downstream receptor sites. While PVN OXT was not found to respond differently to either novelty or kinship status, it is worth noting that this cell population has axonal projections that mediate important processes in the discrimination of social stimuli.⁵⁰ Therefore, to further examine whether PVN OXT is involved in differentiating between conspecific types, future studies could target specific subpopulations of OXT neurons, such as those that project to the amygdala and hippocampus.

AUTHOR CONTRIBUTIONS

Christian Janmil Esquilin-Rodriguez: Investigation; writing – original draft; methodology; writing – review and editing; formal analysis; visualization; data curation. **Brandon A. Fricker:** Conceptualization; investigation; methodology; writing – review and editing. **Aubrey M. Kelly:** Conceptualization; funding acquisition; writing – review and editing; project administration; supervision; resources.

ACKNOWLEDGEMENTS

This work was supported by the National Science Foundation (IOS-2310626 to A.M.K.). B.A.F. was supported by the Department of Defense through the National Defense Science and Engineering Graduate Fellowship Program.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/jne.70089>.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

- Fricker BA, Ho D, Seifert AW, Kelly AM. Biased brain and behavioral responses towards kin in males of a communally breeding species. *Sci Rep*. 2023;13:17040.
- Clemens AM, Wang H, Brecht M. The lateral septum mediates kinship behavior in the rat. *Nat Commun*. 2020;11:3161.
- Rashid M, Thomas S, Isaac J, et al. A ventral hippocampal-lateral septum pathway regulates social novelty preference. doi:10.7554/eLife.97259.1
- Ferguson JN, Young LJ, Hearn EF, Matzuk MM, Insel TR, Winslow JT. Social amnesia in mice lacking the oxytocin gene. *Nat Genet*. 2000;25:284-288.
- Macbeth AH, Lee H-J, Edds J, Young WS. Oxytocin and the oxytocin receptor underlie intraspecific, but not interstrain, social recognition. *Genes Brain Behav*. 2009;8:558-567.
- Ferguson JN, Aldag JM, Insel TR, Young LJ. Oxytocin in the medial amygdala is essential for social recognition in the mouse. *J Neurosci*. 2001;21:8278-8285.
- Raam T, McAvoy KM, Besnard A, Veenema AH, Sahay A. Hippocampal oxytocin receptors are necessary for discrimination of social stimuli. *Nat Commun*. 2017;8:2001.
- Oettl L-L, Ravi N, Schneider M, et al. Oxytocin enhances social recognition by modulating cortical control of early olfactory processing. *Neuron*. 2016;90:609-621.
- Dumais KM, Alonso AG, Immormino MA, Bredewold R, Veenema AH. Involvement of the oxytocin system in the bed nucleus of the stria terminalis in the sex-specific regulation of social recognition. *Psychoneuroendocrinology*. 2016;64:79-88.
- Fischer-Shofty M, Levkovitz Y, Shamay-Tsoory SG. Oxytocin facilitates accurate perception of competition in men and kinship in women. *Soc Cogn Affect Neurosci*. 2013;8:313-317.
- DiBenedictis BT, Nussbaum ER, Cheung HK, Veenema AH. Quantitative mapping reveals age and sex differences in vasopressin, but not oxytocin, immunoreactivity in the rat social behavior neural network. *J Comp Neurol*. 2017;525:2549-2570.
- Kelly AM, Seifert AW. Distribution of vasopressin and oxytocin neurons in the basal forebrain and midbrain of spiny mice (*Acomys cahirinus*). *Neuroscience*. 2021;468:16-28.
- Kelly AM, Goodson JL. Social functions of individual vasopressin-oxytocin cell groups in vertebrates: what do we really know? *Front Neuroendocrinol*. 2014;35:512-529.
- Landgraf R, Neumann ID. Vasopressin and oxytocin release within the brain: a dynamic concept of multiple and variable modes of neuropeptide communication. *Front Neuroendocrinol*. 2004;25:150-176.
- Engelmann M, Landgraf R, Wotjak CT. The hypothalamic-neurohypophyseal system regulates the hypothalamic-pituitary-adrenal axis under stress: an old concept revisited. *Front Neuroendocrinol*. 2004;25:132-149.
- Ludwig M, Leng G. Dendritic peptide release and peptide-dependent behaviours. *Nat Rev Neurosci*. 2006;7:126-136.
- Resendez SL, Nambodiri VMK, Otis JM, et al. Social stimuli induce activation of oxytocin neurons within the paraventricular nucleus of the hypothalamus to promote social behavior in male mice. *J Neurosci*. 2020;40:2282-2295.
- Ross HE, Cole CD, Smith Y, et al. Characterization of the oxytocin system regulating affiliative behavior in female prairie voles. *Neuroscience*. 2009;162:892-903.
- Thirumara Rajamani K, Barbier M, Lefevre A, et al. Oxytocin activity in the paraventricular and supramammillary nuclei of the hypothalamus is essential for social recognition memory in rats. *Mol Psychiatry*. 2024;29:412-424.
- Harmon AC, Huhman KL, Moore TO, Albers HE. Oxytocin inhibits aggression in female Syrian hamsters. *J Neuroendocrinol*. 2002;14:963-969.
- Duque-Wilckens N, Torres LY, Yokoyama S, et al. Extrahypothalamic oxytocin neurons drive stress-induced social vigilance and avoidance. *Proc Natl Acad Sci*. 2020;117:26406-26413.
- Haughton CL, Gawriluk TR, Seifert AW. The biology and husbandry of the African spiny mouse (*Acomys cahirinus*) and the research uses of a laboratory colony. *J Am Assoc Lab Anim Sci*. 2016;55:9-17.
- Gonzalez Abreu JA, Rosenberg AE, Fricker BA, Wallace KJ, Seifert AW, Kelly AM. Species-typical group size differentially influences social reward neural circuitry during nonreproductive social interactions. *iScience*. 2022;25:104230.
- Fricker BA, Murugan M, Seifert AW, Kelly AM. Cingulate to septal circuitry facilitates the preference to affiliate with large peer groups. *Curr Biol*. 2024;34:4452-4463.e4.
- Čížková B, Šumbera R, Frynta D. A new member or an intruder: how do Sinai spiny mouse (*Acomys dimidiatus*) families respond to a male newcomer? *Behaviour*. 2011;148:889-908.
- Shargal E, Kronfeld-Schor N, Dayan T. Population biology and spatial relationships of coexisting spiny mice (*Acomys*) in Israel. *J Mammal*. 2000;81:1046-1052.
- Shkolnik A, Borut A. Temperature and water relations in two species of spiny mice (*Acomys*). *J Mammal*. 1969;50:245.
- Porter RH, Matochik JA, Makin JW. Evidence for phenotype matching in spiny mice (*Acomys cahirinus*). *Anim Behav*. 1983;31:978-984.
- Breed MD. Kin and nestmate recognition: the influence of W. D. Hamilton on 50 years of research. *Anim Behav*. 2014;92:271-279.
- Fricker BA, Seifert AW, Kelly AM. Characterization of social behavior in the spiny mouse, *Acomys cahirinus*. *Ethology*. 2022;128:26-40.
- Rizzi-Wise CA, Wang DV. Putting together pieces of the lateral septum: multifaceted functions and its neural pathways. *eNeuro*. 2021;8:ENEURO.0315-21.2021.
- Friard O, Gamba M. BORIS: a free, versatile open-source event-logging software for video/audio coding and live observations. *Methods in Ecology and Evolution*. 2016;7:1325-1330.
- Paxinos G, Franklin K. *Paxinos and Franklin's the Mouse Brain Stereotaxic Coordinates*. 4th ed. Academic Press; 2012.
- Schindelin J, Arganda-Carreras I, Frise E, et al. Fiji: an open-source platform for biological-image analysis. *Nat Methods*. 2012;9:676-682.
- Pro-Sistiaga P, Mohedano-Moriano A, On IU-B, et al. Convergence of olfactory and vomeronasal projections in the rat basal telencephalon. *J Comp Neurol*. 2007;504:346-362.
- Hurlemann R, Grinevich V, eds. *Behavioral Pharmacology of Neuropeptides: Oxytocin*. Springer International Publishing; 2018. doi:10.1007/978-3-319-63739-6
- Grievich V, Stoop R. Interplay between oxytocin and sensory systems in the orchestration of socio-emotional behaviors. *Neuron*. 2018;99:887-904.
- Martinez LA, Albers HE, Petrulis A. Blocking oxytocin receptors inhibits vaginal marking to male odors in female Syrian hamsters. *Physiol Behav*. 2010;101(5):685-692. doi:10.1016/j.physbeh.2010.08.007

39. Seifert AW, Kiama SG, Seifert MG, Goheen JR, Palmer TM, Maden M. Skin shedding and tissue regeneration in African spiny mice (*Acomys*). *Nature*. 2012;489(7417):561-565. doi:[10.1038/nature11499](https://doi.org/10.1038/nature11499)
40. Haim A, Shanas U, Oates KM, et al. 5-Aminolevulinate synthase is present, but porphyrins do not accumulate, in the Harderian glands of the mole rat, *Spalax Ehrenbergi*, and two species of spiny mice, *Acomys Cahirinus* and *a. Russatus*. *Isr J Ecol Evol*. 1997;43:267-272.
41. Halpin ZT. Individual differences in the biological odors of the mongolian gerbil (*Meriones unguiculatus*). *Behav Biol*. 1974;11:253-259.
42. Thiessen DD, Clency A, Goodwin M. Harderian gland pheromone in the Mongolian gerbil *Meriones unguiculatus*. *J Chem Ecol*. 1976;2(2):231-238.
43. Steinman MQ, Duque-Wilckens N, Greenberg GD, et al. Sex-specific effects of stress on oxytocin neurons correspond with responses to intranasal oxytocin. *Biol Psychiatry*. 2016;80:406-414.
44. Neumann ID, Slattery DA. Oxytocin in general anxiety and social fear: a translational approach. *Biol Psychiatry*. 2016;79:213-221.
45. Smith AS, Wang Z. Hypothalamic oxytocin mediates social buffering of the stress response. *Biol Psychiatry*. 2014;76:281-288.
46. Liao P-Y, Chiu Y-M, Yu J-H, Chen SK. Mapping central projection of oxytocin neurons in unmated mice using Cre and alkaline phosphatase reporter. *Front Neuroanat*. 2020;14:559402.
47. Yao S, Bergan J, Lanjuin A, Dulac C. Oxytocin signaling in the medial amygdala is required for sex discrimination of social cues. *Elife*. 2017;6:e31373.
48. Zhan S, Qi Z, Cai F, Gao Z, Xie J, Hu J. Oxytocin neurons mediate stress-induced social memory impairment. *Curr Biol*. 2024;34:36-45.e4.
49. Kelly AM, Hiura LC, Saunders AG, Ophir AG. Oxytocin neurons exhibit extensive functional plasticity due to offspring age in mothers and fathers. *Integr Comp Biol*. 2017;57:603-618.
50. Rigney N, De Vries GJ, Petrulis A, et al. Oxytocin, vasopressin, and social behavior: from neural circuits to clinical opportunities. *Endocrinology*. 2022;163:bqac111.

How to cite this article: Esquilin-Rodriguez CJ, Fricker BA, Kelly AM. Oxytocin neural responses distinguish social novelty from familiarity but not kin from non-kin in male spiny mice. *J Neuroendocrinol*. 2025;37(12):e70089. doi:[10.1111/jne.70089](https://doi.org/10.1111/jne.70089)